

DECISION TREE

AIM

To develop a Python program to **implement a Decision Tree algorithm** for classification and predict the class of a given input based on decision rules.

OBJECTIVE

- To understand the concept of **Decision Tree** in Artificial Intelligence and Machine Learning.
- To construct a decision tree from training data.
- To classify new data using the constructed tree.
- To understand how attributes are selected for splitting.
- To predict the output based on the decision rules.

ALGORITHM

Step 1: Start.

Step 2: Prepare the training dataset containing attributes and class labels.

Step 3: Select the best attribute for splitting the dataset using **Information Gain**.

Step 4: Divide the dataset according to the selected attribute.

Step 5: Repeat the splitting process for each subset.

Step 6: If all records in a subset belong to the same class, create a leaf node with that class.

Step 7: Continue until the complete decision tree is constructed.

Step 8: Give a new input to the decision tree.

Step 9: Traverse the tree according to the input attributes.

Step 10: Display the predicted class.

Step 11: Stop.

PYTHON SOURCE CODE

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn import tree
```

```
# Training data
```

```
# [Age, Income]
```

```
X = [
```

```
[25, 30],  
[30, 40],  
[35, 50],  
[40, 60],  
[45, 70],  
[50, 80]  
]
```

```
# Class labels
```

```
# 0 = No
```

```
# 1 = Yes
```

```
y = [0, 0, 1, 1, 1, 1]
```

```
# Create Decision Tree
```

```
model = DecisionTreeClassifier(criterion="entropy")
```

```
# Train the model
```

```
model.fit(X, y)
```

```
# Test data
```

```
age = int(input("Enter Age: "))
```

```
income = int(input("Enter Income: "))
```

```
# Prediction
```

```
prediction = model.predict([[age, income]])
```

```
if prediction[0] == 1:
```

```
    print("Prediction: Yes")
```

```
else:
```

```
    print("Prediction: No")
```

```
# Display tree structure
```

```
print("\nDecision Tree:")
```

```
print(tree.export_text(model, feature_names=["Age", "Income"]))
```

OUTPUT

```
===== RESTART: C:/Users/1
Enter Age: 32
Enter Income: 45
Prediction: No

Decision Tree:
|--- Age <= 32.50
|   |--- class: 0
|--- Age > 32.50
|   |--- class: 1
.>
```

RESULT

The Python program successfully implements a **Decision Tree classifier**. The model is trained using the given dataset and predicts the class of a new input based on the decision rules learned from the training data.